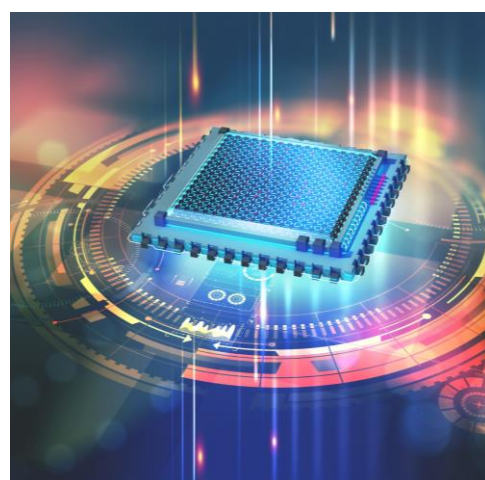
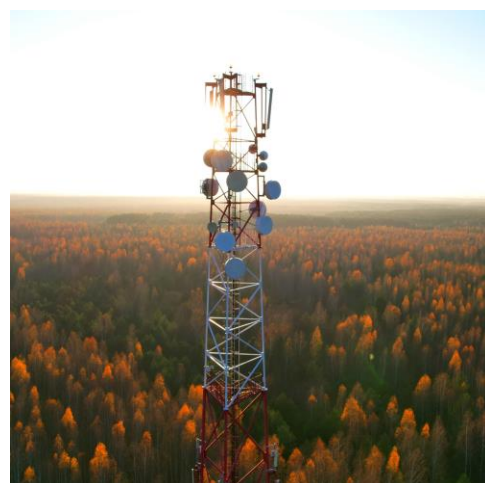


Model 866-M Compact Frequency Synthesizer



Features

- Small size flange-mount module
- CW and pulse modulation
- Low power consumption as low as 23 W, passive cooling

Applications

- System clock source
- Multi-channel phase-coherent configurations for radar signal generation and testing
- Spectroscopy



Model 866-M

1 MHz to 40 GHz Compact Frequency Synthesizer

CONTENTS

CONTENTS	2
DEFINITIONS	3
INTRODUCTION	3
FACTS & FIGURES & SPECIFICATIONS	4
Signal Specifications	4
Frequency Reference	4
Level Performance	5
Level Uncertainty	5
Reverse Power Protection and VSWR	6
Phase Noise	6
Spectral Purity	7
Modulation Capability	8
Sweeping Capability	9
Trigger (TRIG)	9
MECHANICAL SPECIFICATIONS	10
Dimensions & Weight	10
Installation Instructions	10
CONNECTORS	11
Front Panel	11
Rear Panel	11
Power Connector Assembly	12
Reset Functionality	12
ORDERING INFORMATION	13
GENERAL CHARACTERISTICS	13
Document History	14
NOTES	15

DEFINITIONS

- The specifications in the following pages describe the warranted performance of the instrument for $23 \pm 5^\circ\text{C}$ after a 30-minute warm-up period (unless otherwise stated).

Min/Max: Parameter range that is guaranteed by product design, and/or production tested. Warranted performance specifications include guard-bands to account for the expected statistical performance distribution, measurement uncertainties, and changes in performance due to environmental conditions.

Typical: Expected mean values, not warranted performance.

INTRODUCTION

- The Model 866-M is a compact-size frequency synthesizer for the generation of accurate and stable frequency signals in both CW and pulse form, covering a frequency range of 1 MHzs to 40 GHz, with a fast-switching time of 50 μs , an output power of -20 to +25 dBm. The device shows excellent phase noise performance, e.g., at 10 GHz and 20 kHz offset -127 dBc/Hz is measured.

The synthesizer is in a very compact flange-mountable form of 6.5 x 2.4 x 1 inches [165 x 60 x 26 mm], weighs less than 1.3 lbs, and consumes less than 23 W and is only passively cooled. The Ethernet communication port is used for remote connection to a PC for control over GUI software using standard SCPI commands.

The module can receive and provide external references of 100 MHz or 1 GHz. Multiple units can be phase synchronized with one unit acting as the master and the others as slaves, to implement multi-channel PHASE-COHERENT outputs.

The product is suitable for many applications: as low phase noise system clock or reference sources, in the multi-channel phase-coherent configurations for radar, EW, quantum computing, etc.

FACTS & FIGURES & SPECIFICATIONS

Signal Specifications

PARAMETER	MIN	TYPICAL	MAX	NOTE
Frequency Range	1 MHz		40 GHz	
Frequency Resolution		0.001 Hz		GUI SW setting resolution
Phase Adjustment Range	0 deg		360 deg	
Phase Resolution		0.1 deg		
Switching Speed		500 μ s 50 μ s	85 μ s	Option FS

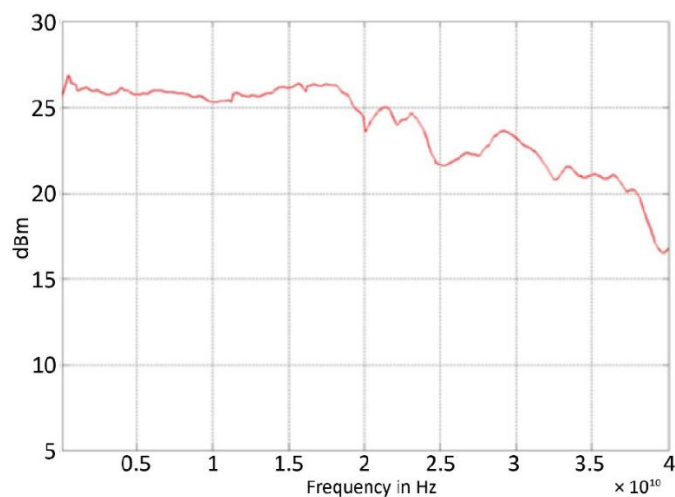
Frequency Reference

PARAMETER	MIN	TYPICAL	MAX	NOTE
Internal reference frequency		100 MHz		
Calibrated accuracy of int. reference		± 30 ppb		Calibrated at 23 ± 3 °C
Temperature stability (0 to 50 °C)			± 100 ppb	
Aging 1 st year			500 ppb	
Aging per day			5 ppb	After 30 days operation
Warm-Up time		5 min		
Reference frequency input	100 MHz, 1 GHz			
Reference input level				
100 MHz	-5 dBm		+10 dBm	
1 GHz	-5 dBm		+10 dBm	
Reference input impedance		50 Ω		
Reference frequency output	100 MHz, 1 GHz			100 MHz reference out not possible when using 1 GHz external reference
Output power				
100 MHz	0 dBm		5 dBm	
1 GHz	-1 dBm		5 dBm	
Reference output impedance		50 Ω		
Multi-Channel configurability				
Recommended configuration	One master and multiple slave units are synchronized by the common 1 GHz reference from the master unit			
Relative phase stability among the multi-channels (phase coherence)		15 mrad		@ 5 GHz over 5 hours

Level Performance

PARAMETER	MIN	TYPICAL	MAX	NOTE
Output power level				See plot below
< 1 GHz	-10 dBm		24 dBm	
1 GHz to 18 GHz	-10 dBm		25 dBm	
18 GHz to 23 GHz	-10 dBm		23 dBm	
23 GHz to 32 GHz	-20 dBm		21 dBm	
32 GHz to 40 GHz	-20 dBm		17 dBm	
Power resolution		0.5 dB		

Figure 1: Typical Maximum Output Power



Level Uncertainty

PARAMETER	MIN	TYPICAL	MAX	NOTE
Power level uncertainty				See plots below
<15 GHz		0.5 dB	1.0 dB	0 to 15 dBm
15 GHz to 35 GHz		1.0 dB	1.5 dB	0 to 15 dBm
35 GHz to 40 GHz		1.0 dB	2.0 dB	0 to 15 dBm

Figure 2: Power Level Linearity

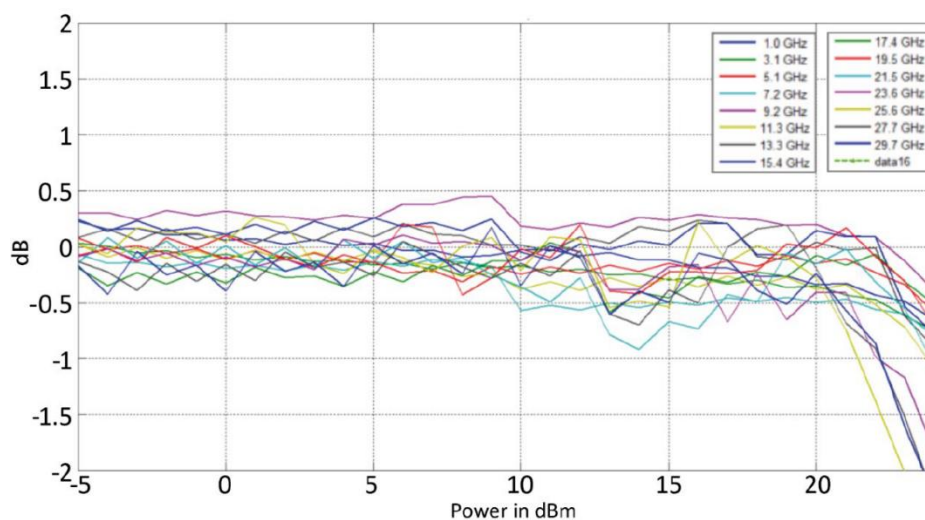
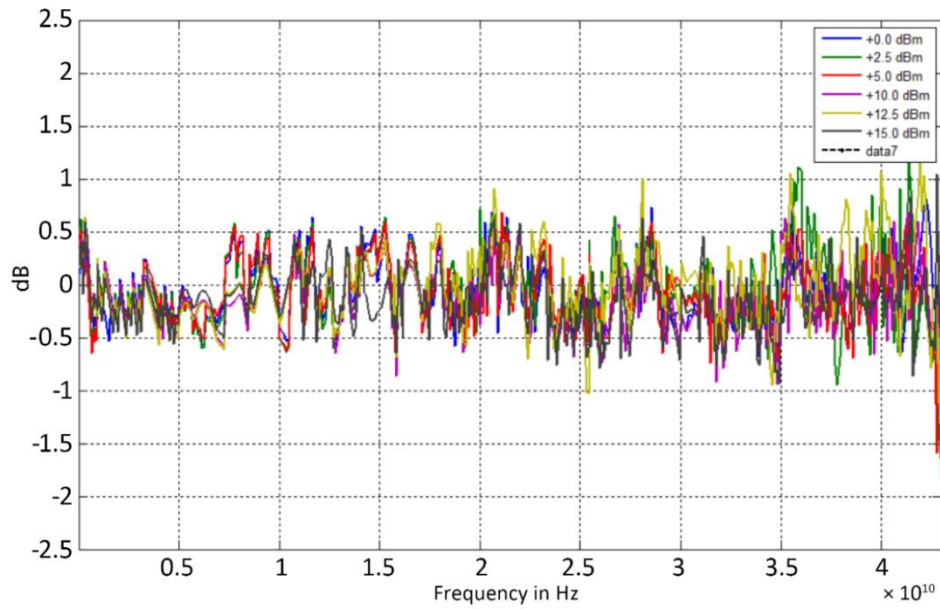


Figure 3: Frequency Response



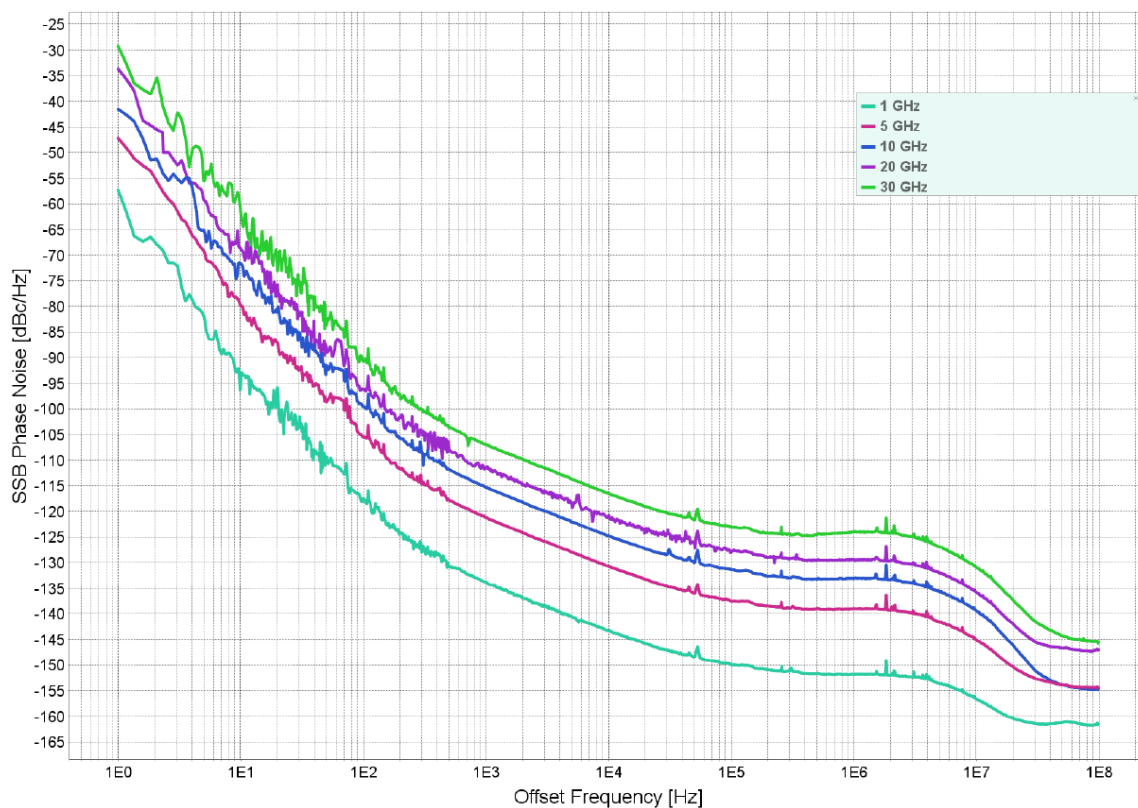
Reverse Power Protection and VSWR

PARAMETER	MIN	TYPICAL	MAX	NOTE
Output impedance		50 Ω		
VSWR		1.7		
Reverse power protection				
DC voltage			7 V	
RF power			23 dBm	

Phase Noise

PARAMETER	MIN	TYPICAL	MAX	NOTE
SSB Phase noise at 1 GHz				See plot below
at 10 Hz from carrier		-87 dBc/Hz	-80 dBc/Hz	
at 1 kHz from carrier		-130 dBc/Hz	-125 dBc/Hz	
at 100 kHz from carrier		-145 dBc/Hz	-140 dBc/Hz	
at 10 MHz from carrier		-155 dBc/Hz	-150 dBc/Hz	
SSB Phase noise at 10 GHz				See plot below
at 10 Hz from carrier		-66 dBc/Hz	-60 dBc/Hz	
at 1 kHz from carrier		-113 dBc/Hz	-108 dBc/Hz	
at 100 kHz from carrier		-129 dBc/Hz	-124 dBc/Hz	
at 10 MHz from carrier		-138 dBc/Hz	-133 dBc/Hz	

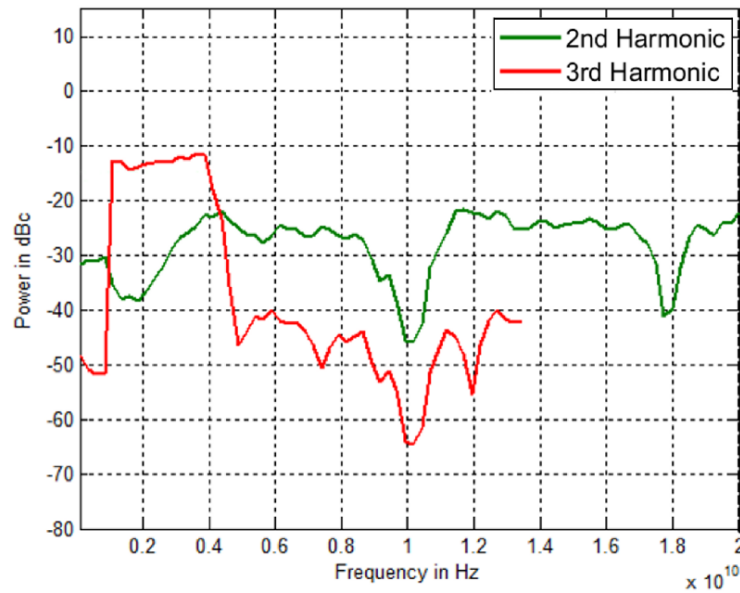
Figure 4: SSB Phase Noise Performance



Spectral Purity

PARAMETER	MIN	TYPICAL	MAX	NOTE
Harmonics				At 10 dBm; See plot below
< 1 GHz		-30 dBc	-25 dBc	
1 GHz to 4 GHz		-15 dBc	-10 dBc	
4 GHz to 20 GHz		-25 dBc	-15 dBc	
20 GHz to 40 GHz		-20 dBc		
Sub-harmonics				At 10 dBm
< 4 GHz		-80 dBc	-70 dBc	
4 GHz to 20 GHz		-70 dBc	-75 dBc	
20 GHz to 40 GHz		-55 dBc	-40 dBc	
Non-harmonic spurious				>10 kHz offset
< 1 GHz		-55 dBc	-50 dBc	
1 GHz to 20 GHz		-70 dBc	-60 dBc	
20 GHz to 40 GHz		-60 dBc		

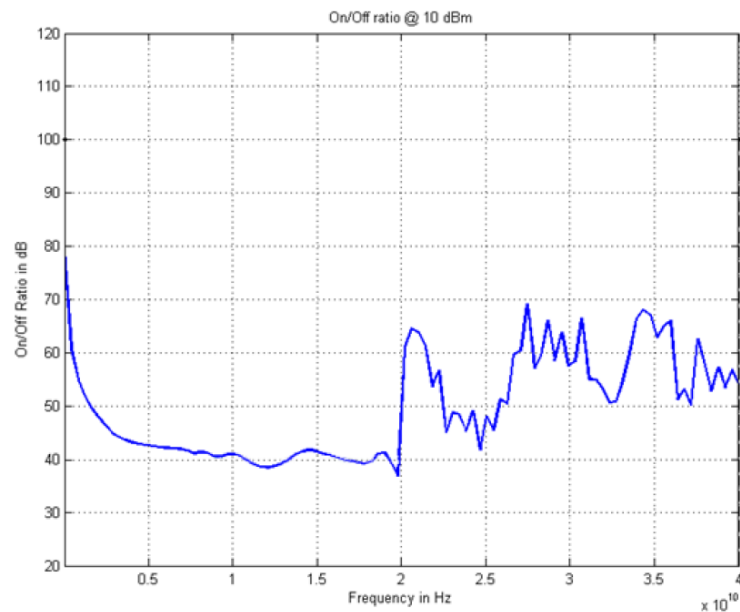
Figure 5: Harmonics (at 10 dBm Output Power)



Modulation Capability

PARAMETER	MIN	TYPICAL	MAX	NOTE
Pulse modulation				
Modulation source	Internal External			Trigger port can be reconfigured as external modulation port
On/off ratio				At 10 dBm; See plot below.
< 1 GHz		60 dB	50 dB	
1 GHz to 20 GHz		40 dB	30 dB	
20 GHz to 27 GHz		45 dB	35 dB	
27 GHz to 40 GHz		55 dB	45 dB	
Repetition frequency	DC		10 MHz	
Pulse width	30 ns		20 s	
Pulse rise/fall time		9 ns		
Pulse train length	1		4096	

Figure 6: On/Off Ratio – Pulse Modulation (at 10 dBm Output Power)



Sweeping Capability

PARAMETER	MIN	TYPICAL	MAX	NOTE
Sweep parameters	Frequency, power, list			
Sweep type	Linear, logarithmic, random			
Step Time	500 μ s 50 μ s		20 s 20 s	Option FS
Timing resolution		10 ns		
Timing accuracy per point		20 ns		
Generalized list sweep				
Allows for individual setting of frequency, power, step-time, and off-time for each point.				

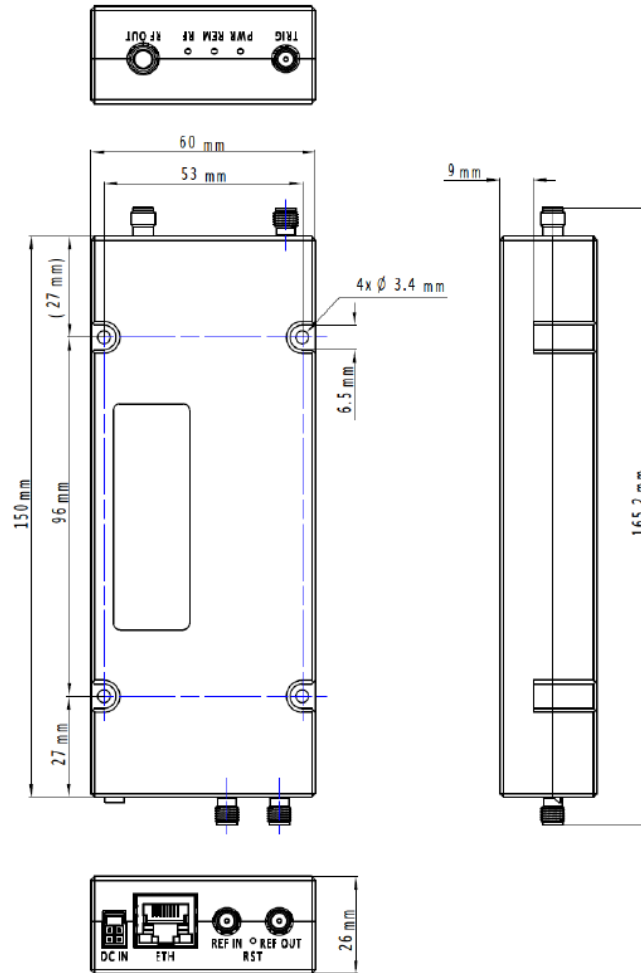
Trigger (TRIG)

PARAMETER	MIN	TYPICAL	MAX	NOTE
Trigger types	Continuous Single (point) Gated Gated direction			
Trigger source	External Ethernet			
Trigger modes	Continuous free run Trigger and run Reset and run			
External trigger latency		140 ns		
External trigger uncertainty		20 ns		
External trigger delay	0 s		20 s	Settable
External delay resolution		10 ns		
Trigger modulo	1		255	Execute only on Nth trigger event
Trigger polarity		Rising Falling		
External trigger input threshold	0.85 V	0.9 V	0.95 V	TTL compatible
External trigger input voltage range	-0.5 V		+5.5 V	TTL compatible
External trigger input hysteresis		60 mV		

MECHANICAL SPECIFICATIONS

Dimensions & Weight

Dimensions / Weight	
Including Connectors	W x L x H = 6.5 x 2.4 x 1" [165 x 60 x 26 mm] / 1 lb [0.45 kg]



Installation Instructions

The module relies on passive cooling. It is mandatory to mount the device on a heatsinking surface. Make sure the synthesizer operates under the conditions specified in the datasheet. Otherwise, the thermal protection will turn off the RF output.

CONNECTORS

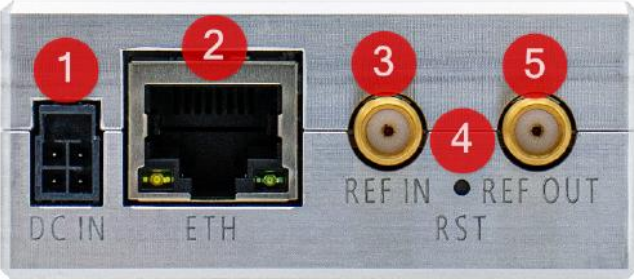
Front Panel

LABEL	TYPE	DESCRIPTION
1. TRIG	SMA	Trigger Signal input
2. PWR	LED	Power ON/OFF indicator
3. REM	LED	Remote connection status indicator
4. RF	LED	RF output ON/OFF indicator
5. RF OUT	2.92 mm	RF output



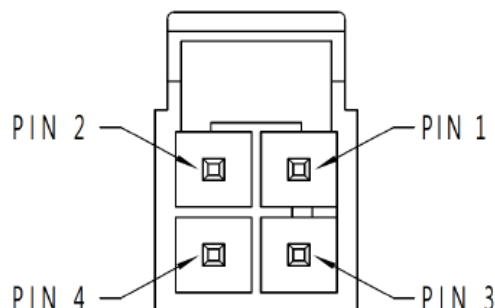
Rear Panel

LABEL	TYPE	DESCRIPTION
1. DC IN	1053141104 (Molex)	DC input (see also chapter “Power Connector Assembly”)
2. ETH	RJ-45	Ethernet port
3. REF IN	SMA	Reference Signal input
4. RST	Button	Reset Button
5. REF OUT	SMA	Reference Signal output



Power Connector Assembly

PIN	ASSIGNMENT
1	DC Supply (see also “Power requirements”)
2	GND
3	DC Supply (see also “Power requirements”)
4	GND



The power connector is a 4 pin, 2.50 mm spaced double-row header. BNC recommends Molex manufactured receptacles 1053081204 and contacts 1053002200 with TPA retainer 1053252002 from its Nano-Fit series.

Reset Functionality

The reset functionality of the device can be used if the device has a wrong configured IP address and cannot be found in the local area network anymore. While pressing the reset button (>2 s), the device will be reconfigured to auto IP addressing mode (DHCP/Zero-conf) and restarts itself. After the device is up and running again, it gets a new IP address and can be discovered by the BNC Signal Generator UI.

ORDERING INFORMATION

HOST MODEL	PRODUCT	DESCRIPTION
866-M	866-M	40 GHz wideband frequency synthesizer module (with AC adapter)
866-M	Option FS	Fast Switching

GENERAL CHARACTERISTICS

Remote Programming Interfaces:

Ethernet

Control language: SCPI Version 1999.0

Power requirements: 24±6 VDC; 23 W maximum

Mains adapter supplied: 100-240 VAC in / 24 V, 2.7 A DC out

Storage temperature range: -40 to 70 °C

Operating temperature range: 0 to 45 °C, non-condensing

Operating and storage altitude: up to 15,000 feet



Safety/EMC complies with applicable Safety and EMC regulations and directives.

Recommended calibration cycle: 24 months



Document History

[illegible]

Berkeley Nucleonics

2955 Kerner Blvd.
San Rafael, CA 94901

Phone: 415-453-9955
Email: info@berkeleyneutronics.com

berkeleynucleonics.com
berkeleynucleonics.com/downloads

NOTES